

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0787

Colfax 100 kV
April 23, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0787. The Interconnection Customer proposes a 146.7 MW wind facility interconnecting at the Colfax 100 kV substation in the Indiana area. The request was received on April 28, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,082,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Imperial / Chmberlin Substation 230.0 kV Line
Contingency: Hylebos / Ravenna 500.0 kV
Loading: 160.3% (Rating: 704 MVA)
DFAX: 10.8%
Pre-project loading: 158.1%
2. Facility: Ravenna / Old Forge 230.0 kV
Contingency: Hylebos / Ravenna 500.0 kV
Loading: 142.5% (Rating: 834 MVA)
DFAX: 8.6%
Pre-project loading: 141.0%
3. Facility: Montpelier Electric Generating Station 500/230kV Transformer
Contingency: Wabash / Montpelier Electric Generating Station 500.0 kV
Loading: 137.5% (Rating: 804 MVA)
DFAX: 7.8%
Pre-project loading: 136.1%
4. Facility: Maple Street / Atascocita 230.0 kV
Contingency: El Paso / Ravenna 500.0 kV L
Loading: 131.0% (Rating: 563 MVA)
DFAX: 5.3%
Pre-project loading: 129.7%
5. Facility: Omni-Hackberry 500-230 kV Autotransformer
Contingency: Omni-Hackberry / Norwich 500.0 kV
Loading: 130.0% (Rating: 942 MVA)
DFAX: 5.6%
Pre-project loading: 129.1%

6. Facility: Omni-Hackberry / Norwich 500.0 kV Line
Contingency: Beaver / Trego 500.0 kV Line
Loading: 125.6% (Rating: 2,332 MVA)
DFAX: 19.2%
Pre-project loading: 124.4%
7. Facility: Canal Rd / Newport Tie 230.0 kV
Contingency: Trego / Beaver 500.0 kV
Loading: 123.5% (Rating: 713 MVA)
DFAX: 10.5%
Pre-project loading: 121.4%
8. Facility: Napoleon Peaking Sta / Minnehaha Substation 230.0 kV
Contingency: Troy Energy Llc / Richland 500.0 kV L
Loading: 122.5% (Rating: 563 MVA)
DFAX: 12.3%
Pre-project loading: 119.3%
9. Facility: Leetonia / Maple Street 230.0 kV
Contingency: El Paso / Ravenna 500.0 kV
Loading: 121.4% (Rating: 594 MVA)
DFAX: 5.3%
Pre-project loading: 120.1%
10. Facility: Ganso Substation / Handsome Lake Substation 500.0 kV
Contingency: Lions Rd / Piney Crk Project 500.0 kV
Loading: 119.9% (Rating: 2,387 MVA)
DFAX: 12.5%
Pre-project loading: 119.1%
11. Facility: Trego / Beaver 500.0 kV
Contingency: Adams / Onancock Diesel Generation Facility 230.0 kV
Loading: 118.1% (Rating: 3,123 MVA)
DFAX: 31.0%
Pre-project loading: 116.6%
12. Facility: Allegheny Energy Units 3 4 & 5 500/1 kV Autotransformer
Contingency: Piney Crk Project / Hilltop 500.0 kV
Loading: 116.4% (Rating: 1,626 MVA)
DFAX: 9.6%
Pre-project loading: 115.6%
13. Facility: Allegheny Energy Units 3 4 & 5 345/1kV TX
Contingency: Hilltop / Piney Crk Project Substation 500.0 kV Line
Loading: 116.4% (Rating: 1,626 MVA)
DFAX: 9.6%
Pre-project loading: 115.6%
14. Facility: Segs 7 Substation / Freehaven-Lakepark 100.0 kV Cir. 2
Contingency: Freehaven-Lakepark / Segs 7 100.0 kV Line 1
Loading: 116.4% (Rating: 199 MVA)
DFAX: 30.8%
Pre-project loading: 93.6%
15. Facility: Caldwell / Blackrock 345.0 kV
Contingency: Carbon Ctr 500/345kV Transformer Bank
Loading: 116.0% (Rating: 1,217 MVA)
DFAX: 5.5%
Pre-project loading: 115.4%
16. Facility: Stewart Mountain Dam / Lighthouse Hill 345.0 kV
Contingency: Piney Crk Project / Hilltop 500.0 kV

- Loading: 115.5% (Rating: 1,059 MVA)
DFAX: 7.4%
Pre-project loading: 114.5%
17. Facility: Piney Crk Project 500-345 kV Transformer Bank
Contingency: Hilltop / Piney Crk Project 500.0 kV
Loading: 115.2% (Rating: 1,923 MVA)
DFAX: 8.6%
Pre-project loading: 114.5%
18. Facility: Lighthouse Hill Substation / Aes Beaver Valley Partners Beaver Vly 345.0 kV line
Contingency: Hilltop / Piney Creek Project 500.0 kV
Loading: 114.9% (Rating: 1,193 MVA)
DFAX: 7.6%
Pre-project loading: 113.9%
19. Facility: Barrett / Norwich 500.0 kV Line
Contingency: Beaver / Trego 500.0 kV
Loading: 114.9% (Rating: 2,554 MVA)
DFAX: 14.1%
Pre-project loading: 114.0%
20. Facility: Suar Hill / Allegheny Energy Units 3 4 & 5 345.0 kV line
Contingency: Hilltop / Piney Crk Project 500.0 kV
Loading: 114.8% (Rating: 1,785 MVA)
DFAX: 9.9%
Pre-project loading: 113.9%
21. Facility: Ganso / El Paso 500.0 kV
Contingency: Lions Rd / Piney Crk Project 500.0 kV
Loading: 114.2% (Rating: 2,779 MVA)
DFAX: 14.0%
Pre-project loading: 113.4%
22. Facility: N Cambridge / Atascocita 230.0 kV line
Contingency: Ravenna / El Paso 500.0 kV line
Loading: 114.0% (Rating: 928 MVA)
DFAX: 9.2%
Pre-project loading: 112.5%
23. Facility: Beaver 500/1kV XFMR
Contingency: Beaver Substation / Eufaula 500.0 kV Line
Loading: 112.5% (Rating: 1,793 MVA)
DFAX: 13.8%
Pre-project loading: 111.4%
24. Facility: Beaver 230-1 kV Auto
Contingency: Eufaula Substation / Beaver 500.0 kV
Loading: 112.5% (Rating: 1,793 MVA)
DFAX: 13.8%
Pre-project loading: 111.4%
25. Facility: Ritter Park Substation / Sugar Hill 345.0 kV Line
Contingency: Hilltop / Piney Crk Project 500.0 kV
Loading: 111.9% (Rating: 1,886 MVA)
DFAX: 10.1%
Pre-project loading: 111.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Imperial / Chamberlin 230.0 kV	Rebuild line with higher-capacity conductor	\$18,029,000	1% (\$196,000)
Old Forge / Ravenna 230.0 kV	Rebuild line with higher-capacity conductor	\$17,974,000	1% (\$191,000)
Montpelier Electric Generating Station 500/230kV Transformer	Replace transformer with higher-rated unit	\$19,353,000	1% (\$154,000)
Atascocita / Maple St 230.0 kV	Rebuild line with higher-capacity conductor	\$44,833,000	1% (\$480,000)
Omni-Hackberry 500/230kV TX	Replace transformer with higher-rated unit	\$30,124,000	1% (\$292,000)
Norwich / Omni-Hackberry 500.0 kV	Rebuild line with higher-capacity conductor	\$64,314,000	1% (\$646,000)
Canal Rd / Newport Tie 230.0 kV	Rebuild line with higher-capacity conductor	\$23,391,000	1% (\$251,000)
Napoleon Peaking Sta / Minnehaha Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$27,245,000	2% (\$397,000)
Leetonia / Maple Street 230.0 kV	Rebuild line with higher-capacity conductor	\$18,522,000	1% (\$198,000)
Ganso / Handsme Lake 500.0 kV	Rebuild line with higher-capacity conductor	\$92,157,000	1% (\$980,000)
Trego / Beaver 500.0 kV	Rebuild line with higher-capacity conductor	\$210,287,000	1% (\$2,290,000)
Allegheny Energy Units 3 4 & 5 500/1 kV Transformer	Replace transformer with higher-rated unit	\$21,687,000	1% (\$230,000)
Allegheny Energy Units 3 4 & 5 345/1kV TX	Replace transformer with higher-rated unit	\$14,066,000	1% (\$149,000)
Segs 7 / Freehaven-Lakepark 100.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,064,000	10% (\$214,000)
Caldwell / Blackrock 345.0 kV	Rebuild line with higher-capacity conductor	\$48,924,000	1% (\$519,000)
Stewart Mountain Dam / Lighthouse Hill 345.0 kV	Rebuild line with higher-capacity conductor	\$22,386,000	1% (\$239,000)
Piney Crk Project 500-345 kV Transformer Bank	Replace transformer with higher-rated unit	\$39,372,000	1% (\$417,000)
Lighthouse Hill Substation / Aes Beaver Valley Partners Beaver Vly 345.0 kV line	Rebuild line with higher-capacity conductor	\$21,901,000	1% (\$234,000)
Barrett / Norwich 500.0 kV Line	Rebuild line with higher-capacity conductor	\$139,682,000	1% (\$1,369,000)
Suar Hill / Allegheny Energy Units 3 4 & 5 345.0 kV line	Rebuild line with higher-capacity conductor	\$21,361,000	1% (\$227,000)
El Paso / Ganso 500.0 kV	Rebuild line with higher-capacity conductor	\$154,775,000	1% (\$1,646,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Cambridge / Atascocita 230.0 kV L	Rebuild line with higher-capacity conductor	\$12,042,000	1% (\$128,000)
Beaver 500/1kV XFMR	Replace transformer with higher-rated unit	\$30,003,000	1% (\$321,000)
Beaver 230-1 kV Auto	Replace transformer with higher-rated unit	\$8,375,000	1% (\$90,000)
Ritter Park Substation / Sugar Hill 345.0 kV Line	Rebuild line with higher-capacity conductor	\$21,166,000	1% (\$224,000)
	Total allocated to project		\$12,082,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.